

AI, Robotics, and Cybersecurity in Healthcare: A Medical Humanities Perspective

A Summary of “The Future of AI and Robotics in Modern Medicine” and “Ethical Cybersecurity Standards for Protecting Patient Data in the Healthcare Digital Age”

By Uzair Khan



Introduction: Technology & Ethics

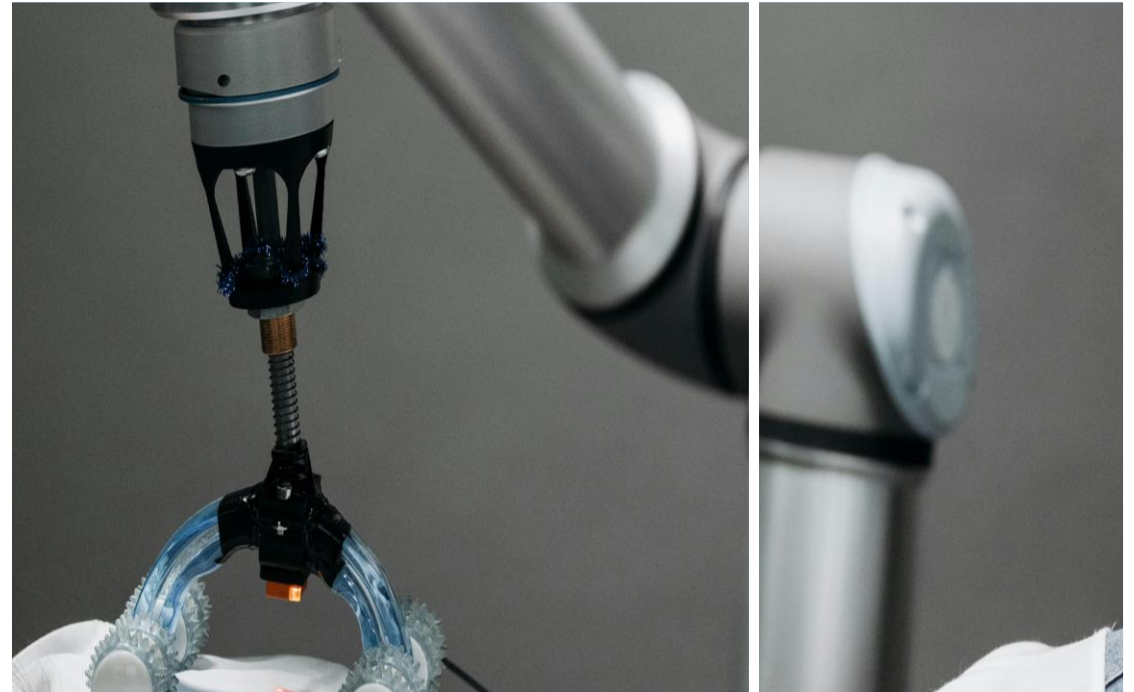
Balancing Innovation and Humanity

- Artificial Intelligence and robotics have “launched a new era in the practice of medicine” by enhancing diagnostic accuracy, tailoring treatments, and improving outcomes.
- Meanwhile, the increased digitalization of healthcare makes patient data vulnerable to cyber threats, raising serious ethical concerns.
- The **medical humanities** serve as a lens to examine how we balance these innovations with empathy, ethics, and patient dignity.

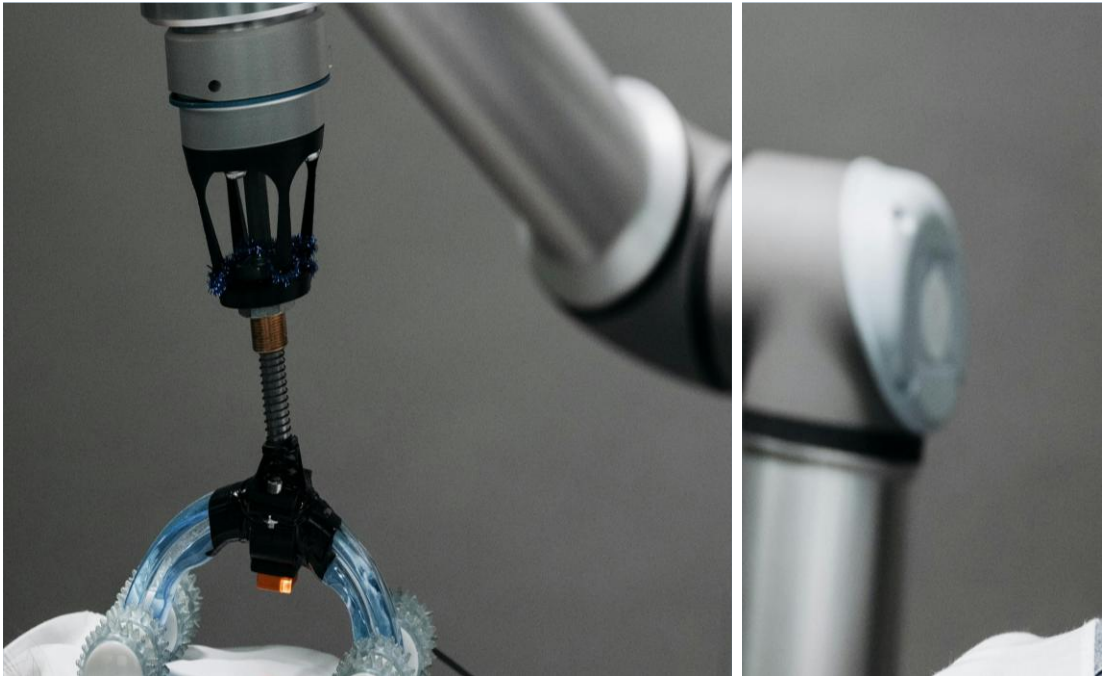


AI in Medicine: Diagnosing with Data

- AI systems “process gigantic volumes of medical information to pick up on subtle patterns no human clinician can see,” enabling early disease detection.
- Studies show AI can surpass physicians in identifying skin cancer and diabetic retinopathy (Esteva et al., 2017; Rajpurkar et al., 2017).
- These tools are not just fast, they are reshaping how clinicians understand illness, risk, and care planning.



Robotic Medicine: New Tools for Healing



- “Robot-assisted surgery reduces recovery time and enhances precision,” helping surgeons perform complex procedures with fewer complications (Makinen et al., 2018).
- In rehabilitation, robotics “help patients regain motor capacity more effectively than traditional methods” (Krebs et al., 2007).
- These technologies minimize physical trauma and accelerate healing but raise questions about human touch in healing.



Dehumanization & Ethical Dilemmas

- “Overdependence on AI platforms can lower empathetic forms of healthcare,” risking emotionally sterile interactions (Char et al., 2018).
- AI decisions “may fail to comply with human ethical guidelines,” leading to questions about accountability and bias (Grote & Berens, 2020).
- The medical humanities urge us to evaluate not only how well AI performs but how ethically it engages with patients.

Case Study: Moxi Robot at Edward Hospital, Naperville

- At Edward Hospital, Moxi robots deliver medication, supplies, and equipment to support overwhelmed and understaffed nurses.
- This case shows how robotics can relieve human burdens and improve workflow during a national shortage.
- Moxi represents a model where AI aids clinicians, rather than erasing their role.



<https://www.nctv17.org/news/moxi-the-robot-arrives-at-napervilles-edward-hospital/>

Cybersecurity in Healthcare: Data Breaches & Privacy

- What was once a profession protected from threat actors is now a high-profile target due to sensitive patient data (Amwins).
- The 2024 Lurie Children's Hospital breach disrupted systems for weeks revealing systemic vulnerabilities.
- Data privacy is not just technical, it is an ethical necessity grounded in autonomy, confidentiality, and non-maleficence (Beauchamp & Childress, 2019).

CHICAGO SUN-TIMES

CRIME NEWS ILLINOIS

800,000 people's data stolen in Lurie Children's Hospital cyberattack

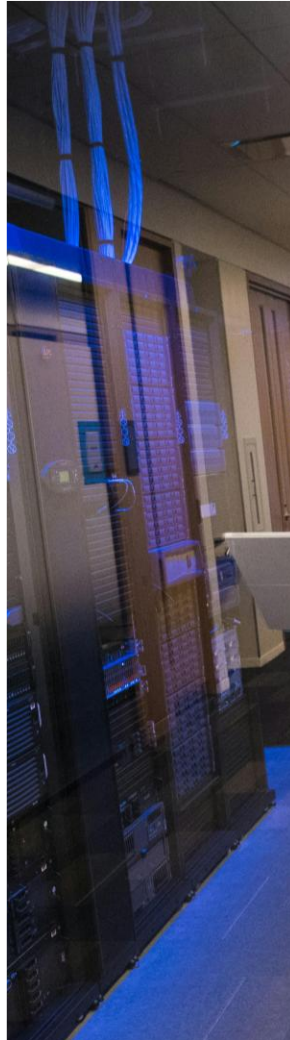
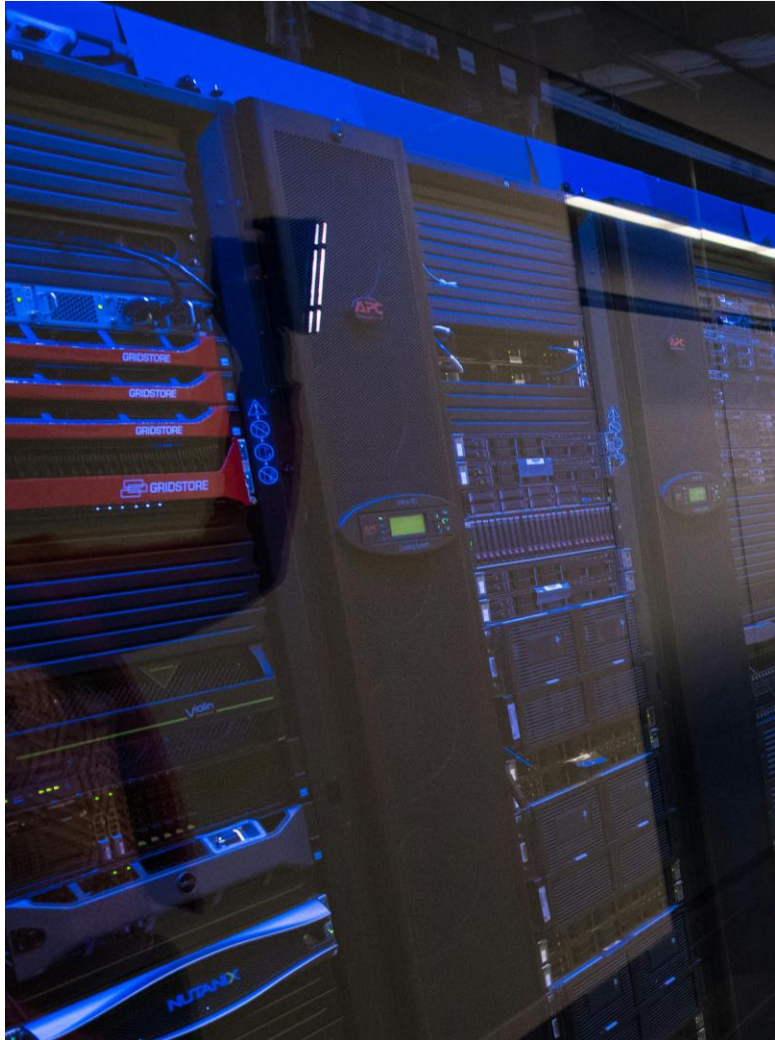
Personal data that was leaked included Social Security numbers, medical conditions or diagnoses, addresses, driver's license numbers and prescription information, the hospital said in a notice on its website.

By Emmanuel Camarillo | Jul 2, 2024, 7:02pm CDT



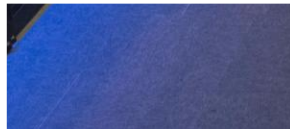
Cybercriminals accessed Lurie's system's between Jan. 26 and Jan. 31, the hospital said. Phone, email and electronic systems were taken offline on Jan. 31 in response to the attack. | Tyler Pasciak LaRiviere/Sun-Times file

<https://chicago.suntimes.com/crime/2024/07/02/lurie-childrens-hospital-cybersecurity-breach-healthcare-tech-data-medical-records-files-fbi>



Securing Healthcare with Cybersecurity Best Practices

- A secure digital system includes:
 - End-to-end encryption
 - Regular system updates
 - Secure data storage
 - Identity and Access Management (IAM) (Venable)
 - Cybersecurity awareness education for professionals
- Healthcare institutions must also adopt transparency and accountability policies for data handling (Kotz et al., 2019).





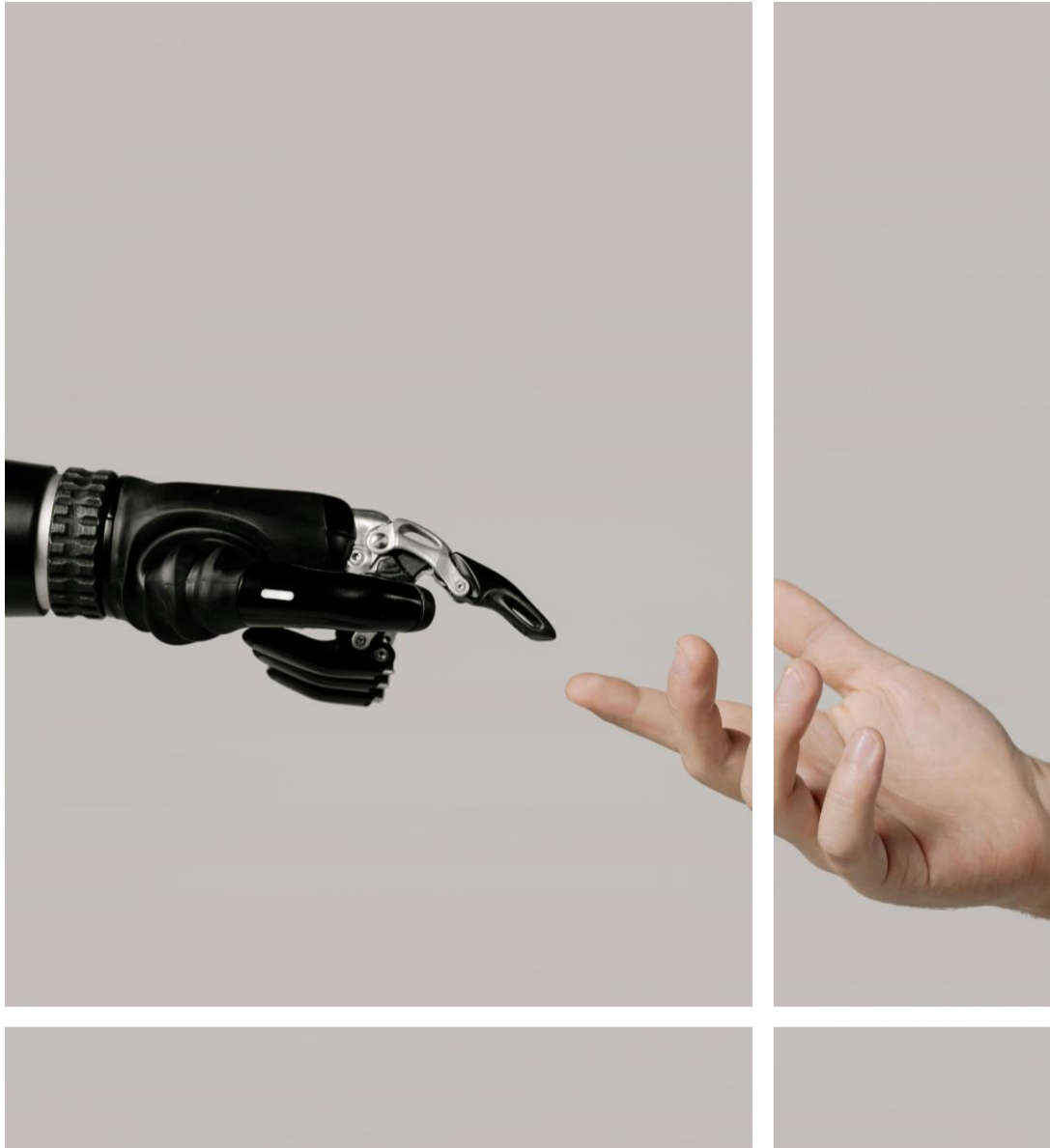
Cybersecurity's Biggest Danger

- “Human error remains one of the greatest vulnerabilities in cybersecurity” (Amwins).
- As a Technology Associate at Staples, I received training on phishing, identity theft, malware, safe passwords, and privacy laws, all of which apply to healthcare.
- Just like medical training, cyber hygiene must be institutionalized in healthcare settings.

Medical Humanities, Patient Rights, and Consent

- Medical humanities promote patient dignity, narrative, and privacy.
- Informed consent, openness in data use, and accountability for data loss are essential to ethical care (Beauchamp & Childress, 2019).
- Healthcare institutions must center patients, not just compliance checkboxes.





The Future of AI and Technology in Healthcare

- As AI evolves, new cyber threats will emerge — including those using AI itself.
- Real-time, AI-driven defense systems will become necessary.
- “Privacy of patient data is an ethical obligation,” not just technical infrastructure (Kotz et al., 2019).
- The future demands both high-tech and high-trust solutions.

Conclusion: Medical Humanities That is Patient-Centered

- Blending AI and robotics with the values of the medical humanities allows for care that is both advanced and patient-centered.
- We must ensure technology remains a tool, not a replacement for empathy, dignity, and narrative-driven care.
- The future of healthcare is not just about what machines can do, but what humans choose to do with them.

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Thank you

